

Damar Albaribin Syamsu

Software Engineer | AI Researcher | System Architect

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PROFESSIONAL SUMMARY

Results-driven Software Engineer with 4+ years of experience in full-stack development, AI/ML research, and system architecture. Proven track record of delivering mission-critical applications across government, telecommunications, fintech, and education sectors. Expert in bridging complex technical execution with business requirements, utilizing modern JavaScript/TypeScript ecosystems, Python AI pipelines, and robust cloud infrastructure (GCP). Known for optimizing application footprints, automating workflows, and leading high-performing engineering teams.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, PHP, C/C++, Dart, Ruby

Frontend: React, Next.js, HTML5, CSS3, Tailwind CSS, Flutter

Backend: Node.js, Laravel, GraphQL, REST APIs, Ruby on Rails

AI & ML: LangChain, RAG Pipelines, YOLO, PyTorch, TensorFlow, Gemini/GPT Integration

Databases: PostgreSQL, MySQL, AlloyDB, Vector DBs (ChromaDB, Pinecone), Redis

DevOps & Cloud: Google Cloud Platform (GCP), Docker, CI/CD, Git, Linux Administration

Other: MAVLink, Arduino, IoT, UAV Systems

PROFESSIONAL EXPERIENCE

Full Stack Developer

Jun 2025 – Jan 2026

Ministry of Environment and Forestry (KLHK), Indonesia – Remote | [Live: conserve-finance](#)

- Architected and deployed the Conserve Accounting APP on GCP, a full-stack dashboard for transparent tracking of USD 6.2 million in UNDP conservation funds.
- Developed robust PostgreSQL databases and RESTful APIs with full audit trail capabilities to ensure international financial compliance.
- Optimized reporting features to reduce manual processing time by 60% through real-time data visualization.

Software Engineer (Backend Focus)

Aug 2025 – Oct 2025

PT Qiscus Tekno Indonesia, Yogyakarta – Internship | [Live: qiscus-shop](#)

- Implemented complex Meta SSO and Meta Business API integrations for a high-traffic e-commerce platform using **Ruby on Rails** and TypeScript.
- Developed robust OAuth 2.0 authentication flows and managed secure API token handling and session orchestration.

AI Researcher

Sep 2024 – Jan 2025

Mobilization Technology B5G/6G Lab, NTUST, Taipei

- Developed AI-driven optimization for 5G/6G network slicing using Transformers and GRUs.
- Achieved an F1-score of 0.9123; optimized resource allocation for eMBB, URLLC, and mMTC.

AI Pipeline Architect

Sep 2024 – Jan 2025

PT Kinema Systrans Multimedia (INFINITE Learning)

- Designed and implemented a YOLOv11 AI pipeline for plant disease detection achieving 95% accuracy.
- Won "Best Product Recognition" among 30+ submissions for the resulting product.

Backend Developer (MSIB)

Feb 2024 – Jun 2024

Widya Robotics, Yogyakarta – On-site

- Developed a full-stack Retrieval-Augmented Generation (RAG) AI application for automated meeting transcription and institutional knowledge management.

Software Engineer & Team Lead

Jan 2022 – Dec 2022

PT Troya Digital Mesail, Yogyakarta

- Architected the Troya Academic Platform using **Laravel**, implementing multi-tenant features and reducing code-base footprint by 99% (100GB to 1GB).
- Spearheaded an automated mobile/web app generator using Node.js and Bash scripting, reducing build cycles from 30 minutes to 60 seconds.
- Led and mentored a team of interns in developing business-focused full-stack features.

Freelance Frontend Developer

Jan 2022 – Jun 2022

PT Eluon Indonesia, Remote | Live: myads.telkomsel.com

- Engineered critical modules for the Telkomsel MyAds V2 platform using Next.js and React, handling national-scale telecommunications traffic.
- Developed USSD Push/Pull and Topup interfaces, consuming complex GraphQL APIs across 7 production sprints.

KEY PROJECTS, COMPETITIONS & AWARDS

Legal & Financial RAG AI Assistants

2024

GEMASTIK XVII (Finalist) & FEKDI Hackathon Bank Indonesia (Finalist)

- Built production-ready AI chatbots utilizing LangChain, GCP Gemini, AlloyDB, and Vector DBs (ChromaDB, Pinecone) for legal and financial compliance domains.
- Achieved 86% retrieval accuracy in law-specific queries against GPT-4/Gemini baselines via advanced RAG pipeline engineering and prompt optimization.
- Represented UNY at two highly competitive national-level competitions simultaneously, advancing to the finalist round in both.

Dreamagination Autonomous UAV System

2023

National Flying Robot Competition (KRTI) – 2nd Place (ESC) & 4th Place (GCS/Flight Controller), 186 participants from 64+ universities

- Served as Lead Engineer & System Architect for a fully autonomous drone system, designing an architecture bridging low-level hardware (serial/I2C, Raspberry Pi) with high-level control software.
- Implemented MAVLink for UAV telemetry and defined a custom communication packet enabling seamless data exchange between the Ground Control Station, ESC, and Flight Controller teams.
- Designed a real-time antenna tracker concept and orchestrated cloud-synchronized program execution across all system components during live competition.
- Extended support to the Electronic Speed Controller (ESC) team by architecting a web-based GUI for real-time ESC monitoring, contributing to their 2nd place finish.

Tanamaku (Agricultural AI)

2024

IBM Academy – Best Product Recognition (among 30+ competing teams)

- Architected an end-to-end AI pipeline for automated tomato disease detection and quality classification, achieving 95% accuracy using a custom-trained YOLOv11 model.
- Applied model quantization and efficient preprocessing pipelines to optimize inference for deployment on resource-constrained mobile hardware.
- Designed a mobile-first interface enabling farmers with minimal technical literacy to diagnose plant health in real-time from field images.

Virtual English Tutor

2022

GEMASTIK XV – 2nd Place, Software Development, Brawijaya University (2,885+ teams from 215 universities nationwide)

- Engineered a conversational AI language learning app powered by **Whisper** (speech recognition), **GPT-3** (adaptive dialogue), and **Live2D** (expressive avatar).
- Personally responsible for building the **RESTful API backend** and the **Progressive Web App (PWA)** frontend serving the full real-time conversational experience.
- Competed against 2,885+ teams from 215 universities — one of Indonesia’s largest national ICT student competitions.

LEADERSHIP & ORGANIZATIONS

Academic Lead | *Google Developer Student Club (GDSC) UNY*

2022 – 2024

Lead Engineer & System Architect | *Dreamagination Robotics Team UNY*

2022 – 2024

IT Coordinator & Backend Lead | *TIM PKM Center UNY* | Live: srgate PKM

2022 – 2024

- Developed and maintained the SRGATE PKM portal using **Laravel**, managing student proposal workflows and data synchronization for the university’s research excellence center.

R&D Staff & Mentor | *INFINITE UNY*

2022 – 2023

EDUCATION

Master of Science in Network Learning Technology <i>National Central University, Taiwan</i>	Jan 2026 – Present
Bachelor of Education in Computer Science (Informatics) <i>Universitas Negeri Yogyakarta, Indonesia</i>	Sep 2021 – May 2025
GPA: 3.79/4.00 Honors: Cum Laude Outstanding Student Award	
Thesis: Deep Learning Architectures for 5G Network Slicing Traffic Classification	
Massive Open Online Course Program (Top 12.7% Acceptance) <i>Universitas Indonesia, Online</i>	Aug 2022 – Dec 2022
Coursework: Infrastructure Management (Grade: B), System Programming (Grade: A)	

LANGUAGES

Indonesian: Native | **English:** Highly Proficient (Professional Working) | **Japanese:** Elementary

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Selected Full-Stack Projects

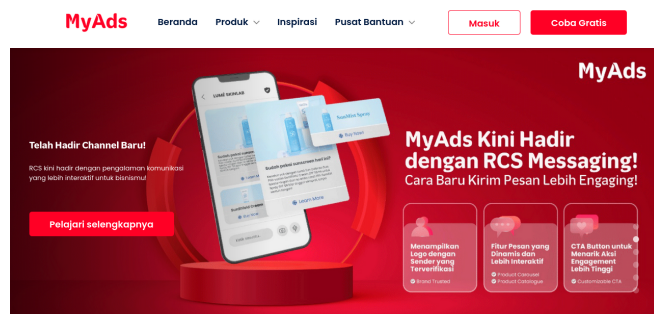
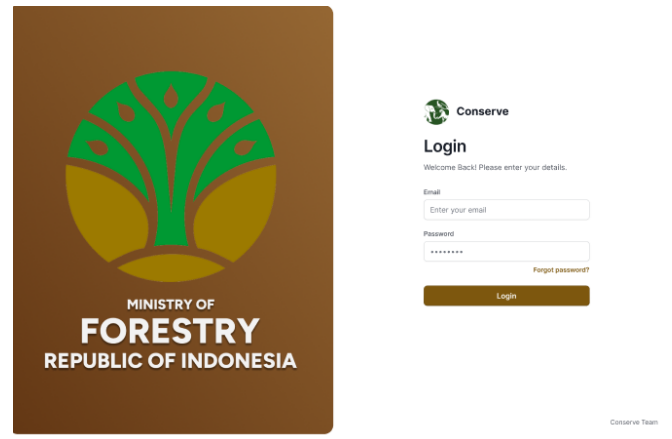
Conserve Finance Dashboard

KLHK & UNDP, Indonesia

Architected and deployed a full-stack dashboard on GCP for transparent tracking of USD 6.2M in conservation funds. Reduced manual reporting time by 60%.

Stack: React, Next.js, PostgreSQL, GCP

Live: ksdae.kehutanan.go.id/conserve-finance



Produk MyAds



Telkomsel MyAds V2

PT Eluon Indonesia

Engineered critical modules including USSD Push/Pull and Topup interfaces for Telkomsel's largest telecom advertising platform, managing national-scale traffic.

Stack: Next.js, React, GraphQL

Live: myads.telkomsel.com

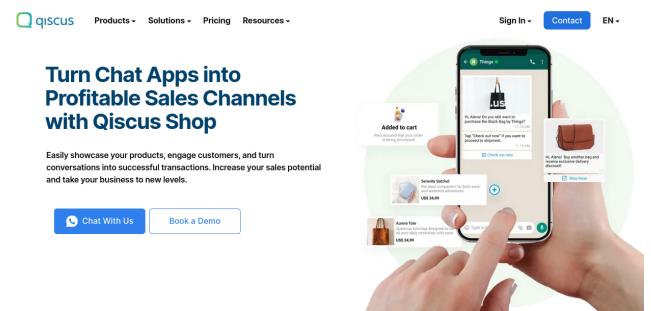
Qiscus Shop Integration

PT Qiscus Tekno Indonesia

Implemented Meta SSO and Meta Business API integrations for a high-traffic e-commerce platform. Developed robust OAuth 2.0 authentication flows.

Stack: Ruby on Rails, TypeScript

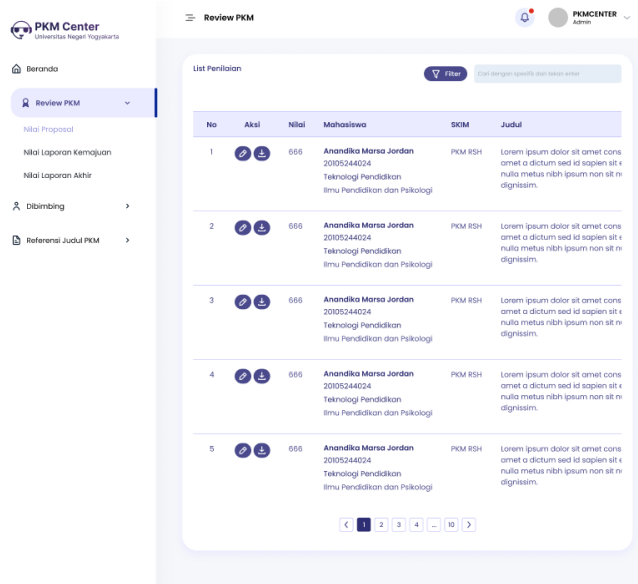
Live: qiscus.com/en/qiscus-shop



Unlock Growth Opportunities of Chat Commerce with Qiscus Solutions

With Qiscus Omnichannel Chat, Qiscus Shop, and AgentLabs, businesses gain access to a wide range of opportunities to enhance the effectiveness of WhatsApp Commerce and other chat applications on various e-commerce platforms.





SRGATE PKM Portal

Universitas Negeri Yogyakarta

Developed and maintained the SRGATE PKM portal managing student proposal workflows and data synchronization for thousands of active users.

Stack: Laravel, PHP, MySQL

Live: srgate.kemahasiswaan.uny.ac.id/pkm

Other Notable Work

Legal AI Chatbot Platform

Built production-ready AI chatbots (RAG) achieving 86% law retrieval accuracy. Finalist at GEMASTIK XVII & FEKDI Hackathon.

Live Preview: youtu.be/16Uh54my7iA

Virtual English Tutor PWA

Conversational AI powered by Whisper (speech), GPT-3 (dialogue), and Live2D. 2nd Place GEMASTIK XV.

Live Preview: youtu.be/g2hooQie7nU

Troya Academic Platform

Multi-tenant SaaS in Laravel. Cut codebase footprint by 99% (100GB to 1GB) and build time down to 60s via automated generator.

(Note: Platform is no longer hosted; led development during Jan '22 – Dec '22)